

```
library("metafor")
```

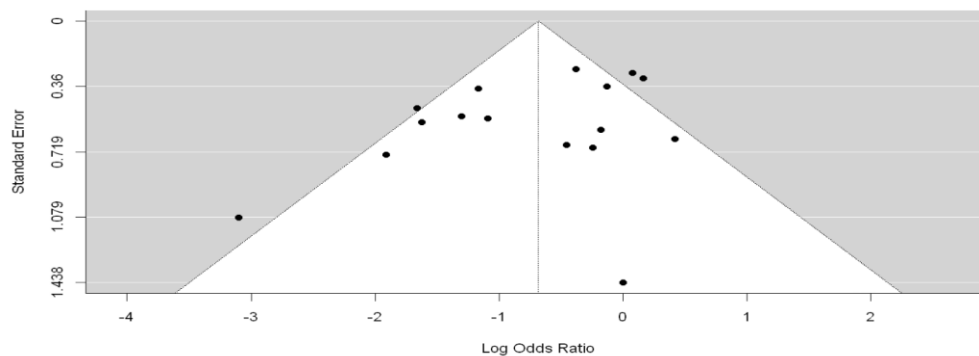
```
data(dat.lee2004)dat.lee2004
```

Effect size: Odds Ratio

```
> dat.lee2004
  id study year ai n1i ci n2i
1  1  Agarwal 2000 18 100 20 100
2  2  Agarwal 2002  5  50 18  50
3  3  Alkaissi 1999  9  20  7  20
4  4  Alkaissi 2002 32 135 31 139
5  5  Allen 1994  9  23 10  23
6  6 Andrzejowski 1996 11  18 12  18
7  7  Duggal 1998 69 122 80 122
8  8  Dundee 1986  3  25 12  25
9  9 Ferrera-Love 1996  1  30  1  30
10 10 Gieron 1993 11  30 19  30
11 11 Harmon 1999  7  44 16  39
12 12 Harmon 2000  4  47  6  47
13 13 Ho 1996  1  30 13  30
14 14 Rusy 2002 24  40 71  80
15 15 Wang 2002 16  50 53  88
16 16 Zarate 2001 28 110 25 111
```

1. [4 pts] Assess for the existence of publication bias:

- using graphical funnel plot
- Egger's regression test



Call:

```
lm(formula = effsizes[, 1]/effsizes[, 2] ~ 1/effsizes[, 2], data = dat)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-5.8580 -2.3233  0.7055  2.1550  4.1459
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -2.5289     0.7668  -3.298  0.00488 **
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 3.067 on 15 degrees of freedom

2. [4 pts] Based on your results above, write your overall conclusion about the existence of publication bias.

The funnel plot appears to be somewhat asymmetrical, with a few studies outside the funnel and not mirrored from each other. This immediately suggests some publication bias, but Eggers test reveals that there is significant bias because the parameter estimate for the intercept in the regression is significant at $\alpha=1\%$.

3. [4 pts] Apply the trim-and-fill method to correct for the publication bias.

Estimated number of missing studies on the right side: 3 (SE = 2.7229)

Equal-Effects Model (k = 19)

I² (total heterogeneity / total variability): 68.84%

H² (total variability / sampling variability): 3.21

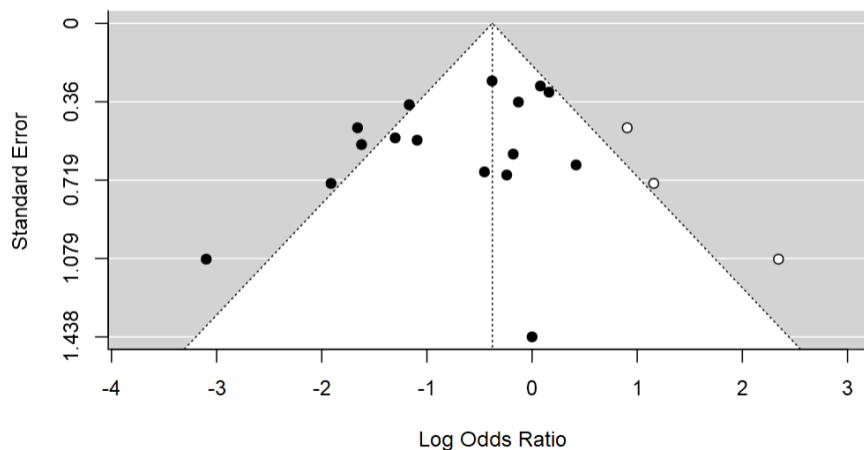
Test for Heterogeneity:

Q(df = 18) = 57.7650, p-val < .0001

Model Results:

estimate	se	zval	pval	ci.lb	ci.ub
-0.3776	0.1080	-3.4960	0.0005	-0.5893	-0.1659

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1



4. [4 pts] Compare your overall effect size estimate before and after adjusting for the publication bias.

The original overall effect size estimate was -0.5145 prior to adjustment. After adjusting for publication bias, the estimate was increased to -0.3776. This occurred because the funnel plot was asymmetric with outliers on the left side, so three studies were imputed on the right side (mirroring the outliers), pulling the pooled effect size estimate up to 0. Both estimates had a similar standard error (0.1128 and 0.108, respectively), and were significant at alpha = 0.1%, although the adjusted estimate was slightly less significant because its magnitude was smaller (as in it was closer to 0, so the null hypothesis that it is different to 0 is inherently more difficult to reject).